UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 7C –Work Sampling

EIN 3241: Human Factors and Ergonomics
Spring 2026

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Admin Issues

- Lab 6 due TONIGHT

Introduction

- Work sampling: technique used to determine the amount of time devoted to the various activities that constitute a job
- Developed by LHC Tippett in 1934 in order to study the operations in the textile industry
- Work sampling: take a large number of observations at random intervals and use that to approximate the desired time
  - What people are doing
  - What machines and equipment are being used
- Work sampling: measurements are made over a long period of time, from days to weeks and months

Examples

Percentage of time a certain machine is used

Idle versus working time of the operator

What percentage of time facilities are used

Pros and Cons

Advantages	Disadvantages
Can determine personnel and machine utilization	Not well suited for short-cycle jobs or short elements
Worker not subjected to long periods of stopwatch observation	Cannot really determine speed of worker or efficiency of work
Crew operations can be studied by a single analyst	Sampling could be biased
Do not need to identify “near-perfect” worker	

Biasing

- Taking observations at times that are synchronous with the operation or that do not give a full picture
  - Always sampling in the morning/evening
  - Always sampling on certain days or at certain times

If you always go early in the morning, a baker will always be very busy.

If you always go at the end of a cycle, the worker might always be idle.

Work sampling steps

1. Plan the study
2. Determine the number of observations needed
3. Determine the frequency of observations
4. Conduct the study
5. Calculate standard time (and/or other desired times)

Remember that you always document the job first

Work sampling steps

1. Plan the study
2. Determine the number of observations needed
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Plan the study

1. Determine the problem: why do you want to conduct this study? To calculate standard time or other times?
2. Determine how data will be collected
   i. Observation: go and come in person
   ii. Video: record a video and analyze later
   iii. Instrumentation: some machines/tools can record what a person is doing
   iv. Self-reports: ask a person to record their actions (could be biased)

Work sampling steps

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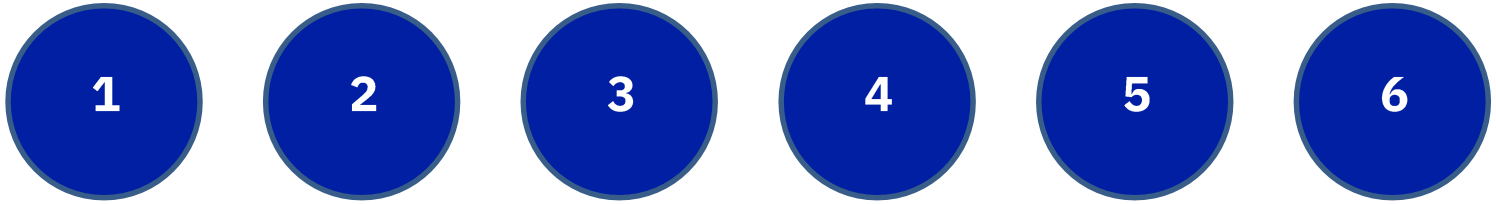
Work sampling theory

- What we are trying to find: percentage of time that a certain task, situation, event occurred
- At any time, this task could be present or absent
- How can one find the probability of this event happening x times out of n samples?



Work sampling theory

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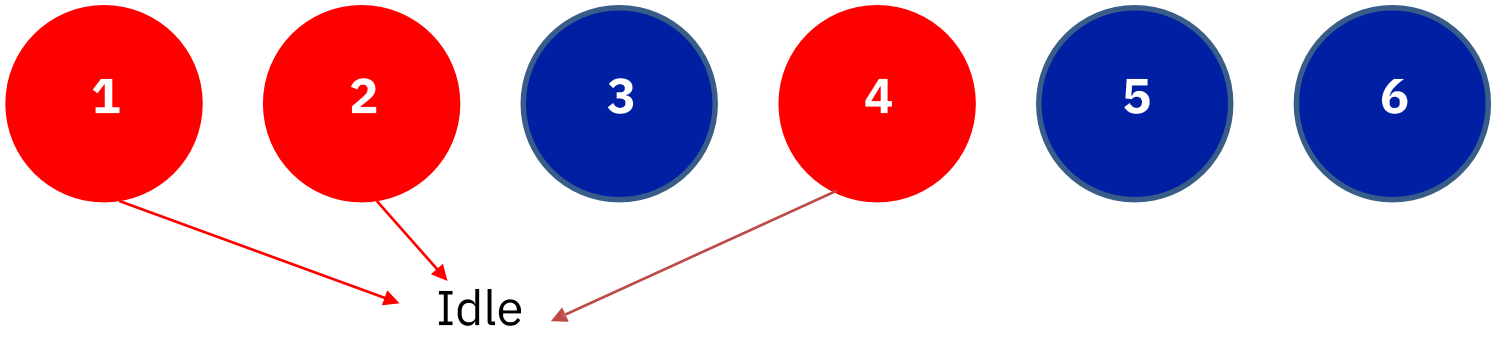


Work sampling theory

- If you checked on a worker 5 times during the day, what is the probability that 3 times out of these 5 checks, that worker will be idle (as opposed to busy)?

$$P(x) = \frac{n!}{x!\,(n-x)!}\,p^x q^{n-x}$$

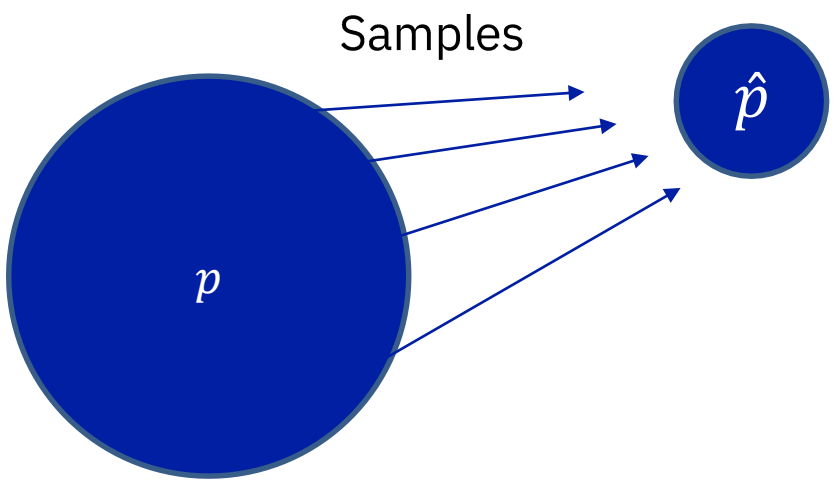
$$p = probability\ of\ a\ single\ occurence$$
$$q = probability\ of\ absence\ of\ occurence = 1 - p$$
$$n = number\ of\ observations$$

$$\boxed{P(3) = \frac{5!}{3!\,(2)!}\,p^3 q^2}$$

- Work sampling is about trying to determine p
- How many times do I need to sample in order to determine p?

Work sampling theory

- Want to estimate p from a certain population



So how many samples do you need?

- For a given number of samples N and number of positive observations x_i for observations

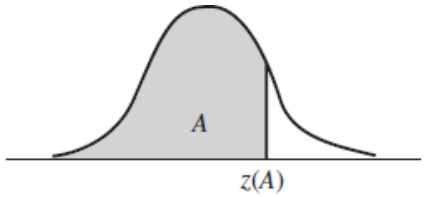
$$\hat{p} = \frac{\sum_1^k x_i}{N}$$

Work sampling theory

- As the sample size increases, can use the normal distribution to approximate the original distribution
- When you sample, $\hat{p}$ will not necessarily be the same as p
  - Ideally, want $\hat{p}$ to be as close as possible
  - Can determine the confidence interval and margin of error for $\hat{p}$ to indicate how close you are to p
- How to find the desired sample size:
  - You have alpha (or can calculate from 1 – confidence level)
  - You have $\hat{p}$, based on estimates or pilot study data
  - Calculate the critical probability $p^* = 1 - \frac{\alpha}{2}$
  - Find the z score in the table with a cumulative probability of p^*
  - Margin of error $= l = Z_{\frac{\alpha}{2}} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
  - $\boxed{n = Z_{\frac{\alpha}{2}}{}^2 \left(\frac{\hat{p}(1-\hat{p})}{l^2}\right)}$

Example 1

- An analyst wishes to determine the amount of downtime due to tool problems in an area involving a machine where very fine drilling is taking place. A study indicated that out of 50 observations, only 2 showed the machine down.
- What would be the number of samples needed for a more accurate study within +/- 1% of the true value and a 99% CL?



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	[illegible]	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823

Selected Percentiles							
Cumulative probability A:	**0.90**	**0.95**	**0.975**	**0.98**	**0.99**	**0.995**	**0.999**
$z(A)$:	1.282	1.645	1.960	2.054	2.326	2.576	3.090

Work sampling steps

1. Plan the study
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5. Calculate standard time (and/or other desired times)

Determine observation frequency

- When should the observations be taken?
- Try to spread out your observations
  - Assume your desired n = 500 samples
  - Assume your available time is 20 days
  - Target (around) 25 observations per day
- What factors affect the observation frequency?
  - Number of analysts available
  - Location of the work
  - Difficulty of access
  - Method of measurement
- Want to avoid biasing: observations synchronous with the process
  - Do not always go at the same times
  - Do not go at fixed intervals for cyclical work

Determine observation frequency

- How to take observations
  - Want to take observations randomly at different times of day
  - Need to include normal fluctuations in production
    - Usually over two to four weeks, *if possible*
- Identify random times (e.g., using Excel)
  - RAND()
  - RAND()*number of hours
  - Convert to time
  - Eliminate times outside of work hours

RAND()	RAND()*8	Format
0.759746	6.077965	~~1:52 AM~~
0.563458	4.507662	12:11 PM
0.580294	4.642351	3:24 PM
0.683156	5.465251	11:09 AM
0.120354	0.962835	~~11:06 PM~~
0.483987	3.871898	~~8:55 PM~~
0.843189	6.745512	~~5:53 PM~~
0.532112	4.256895	~~6:09 AM~~
0.836348	6.69078	4:34 PM
0.049028	0.392223	9:24 AM
0.413224	3.305791	~~7:20 AM~~
0.309182	2.473459	11:21 AM
0.486379	3.891029	~~9:23 PM~~
0.331168	2.649343	3:35 PM
0.455113	3.640905	3:22 PM
0.128721	1.02977	~~12:42 AM~~

Work sampling steps

1. Plan the study
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Conduct the study

- Create the work sampling form

Observations	Time	Using M/C 1	Using M/C 2	Transporting materials	Idle
1	8:30 am		X		
2	9:40 am	X			
3	11:10 am			X	
4	2:15 pm			X	
5	3:00 pm				X
6	3:15 pm		X		
7	4:25 pm		X		

Conduct the study

- Create the work sampling form
- Tips for good recording:
  - Do not anticipate the recording: do not go so that you see what you want to see! You have to be unbiased
  - Go to a fixed point, make the recording
  - Note that arrival of analyst can influence the participant
    - Recall Hawthorne effect: people work better when they are observed

Obs. nos.	Random time	Productive occurrences							Nonproductive occurrences						
		Mch	Weld	Pipe fit	Gen. labor	Elect.	Carpen.	Janitor	Get tools	Grind tools	Wait job	Wait crane	Confer foreman	Personal	Idle
1															
2															
3															
4															
5															
6															
7															
8															
9															
10															
11															
12															
13															
14															
15															
16															
17															
18															
19															
20															
	Total														

Work sampling steps

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Work sampling theory

- For a given number of samples N and number of positive observations x_i (with i from 1 to k)

$$\hat{p} = \frac{\sum_1^k x_i}{N}$$

$$Element\ tme\ (ET) = \frac{T}{P} \times p$$

T=Total time and P=Production units (if applicable)

- If a rating is applied to each element:

$$Normal\ element\ time\ (NT_n) = ET_n \times \frac{R_n}{100}$$

$$Normal\ time = \sum_{i=1}^{n} NT_n$$

- If a rating is applied to the job as a whole:

$$Normal\ time = \sum_{i=1}^{n} ET_n \times \frac{R}{100}$$

Work sampling theory

- Calculation of standard time depends on allowances

$$Standard\ time = NT\ (1 + \frac{Allowance}{100})$$

- Can calculate many other times that do not involve ratings and allowances
  - Machine working time
  - Room utilization time
  - Workstation utilization time

Example 2

- Consider the following results of a pilot work sampling study. With an operator rating of 110% and an allowance of 15%, what is the standard time per unit?

Information	Data
Total working day	480 min
Number of units drilled	420
Working fraction	85%

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LEADING THE CHARGE, CHARGING AHEAD